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A New Lesson Every Sunday
A New Topic Every Month

The Architecture of Linguistic Discretisation

Analysing Tokenisation Strategies, Hidden Flaws, and the Future of Large Language Models.

Tokenisation is the hidden architectural engine of modern AI. The algorithmic mechanism selected to partition text fundamentally dictates a model's embedding scale, attention compute, morphological comprehension, and inherent language biases.

Based on the research of Dr Partha Majumdar, Department of Computer Science, Swiss School of Business and Management (SSBM), Geneva, Switzerland.
(ORCID: 0009-0002-7375-8034).



The Tokenisation Trade-Off: Balancing LLM Performance and Cost

Vocabulary Size: The Performance Trade-off

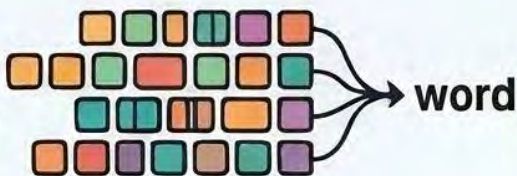
Small Vocab (32k)

Headline: Sequence Compression vs. The Softmax Bottleneck.

Supporting Detail: Large vocabularies process documents faster but slow down the final word generation step.

Token Density

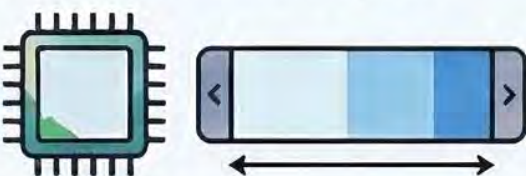
Low (More tokens per word)



Data Table

VRAM Usage:

Lower (More context space)



Training Cost:

Cheaper per step



Large Vocab (128k+)

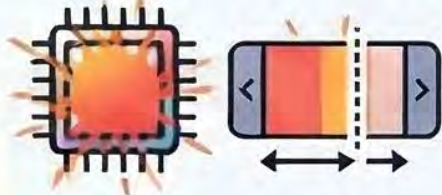
Headline: Eliminating the “Language Tax”.

Supporting Detail: Modern 100k+ vocabularies make non-English processing up to 3x cheaper and more efficient.



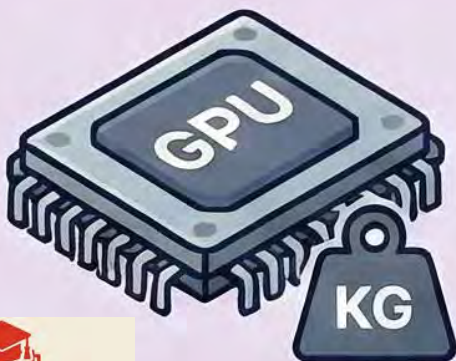
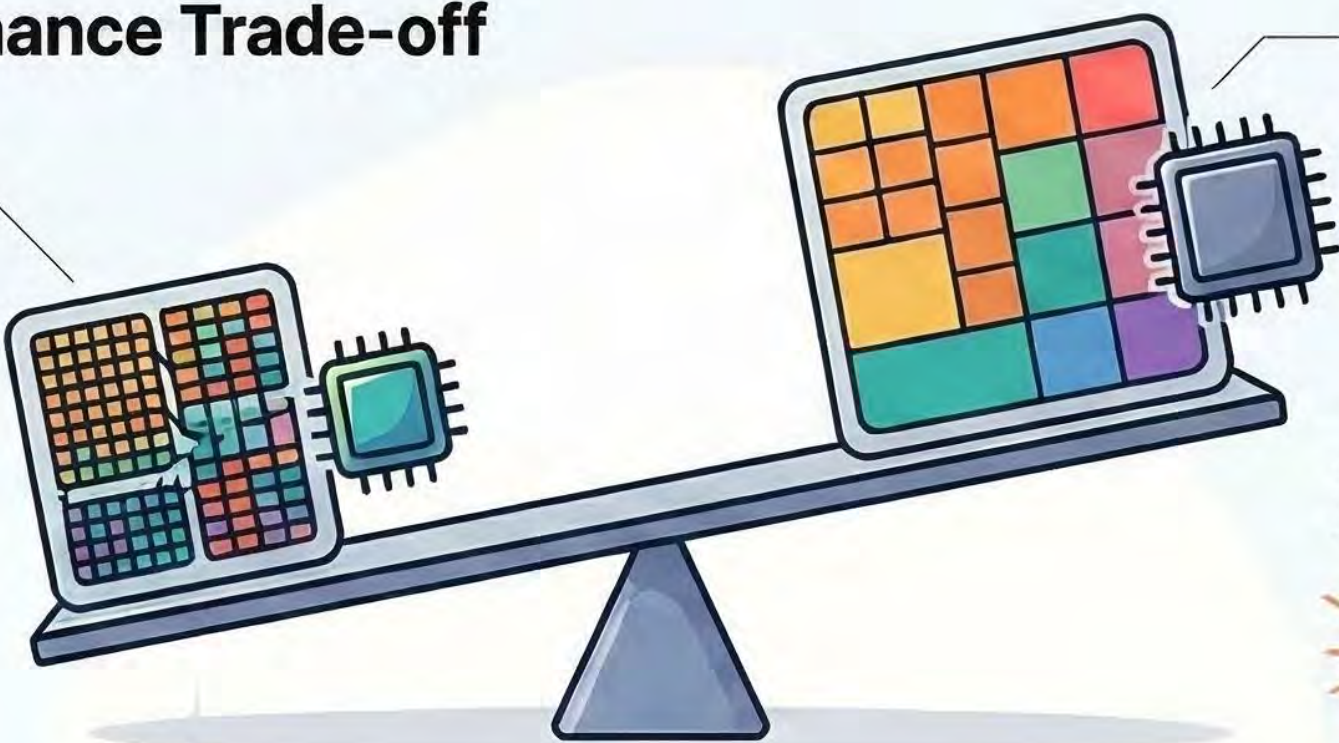
VRAM Usage:

Higher (Limited context space)



Training Cost:

More expensive; needs more data



The VRAM and Parameter “Tax”



Supporting Detail: Larger vocabularies require billions of extra parameters and significantly more GPU memory.

Economic Optimisation in Production



Prompt Caching for Redundancy

Reuse pre-calculated data to get up to 90% discounts on input costs.



50% Flat Discount via Batching

Bundle non-urgent tasks for background processing to cut API costs in half.



Token Density = Real-World Savings

Efficient tokenizers act as a hidden price cut by using fewer tokens.

The Huffman Analogy: Why Vocabularies are Essential

The Huffman Mechanism

INDIA IS GREAT

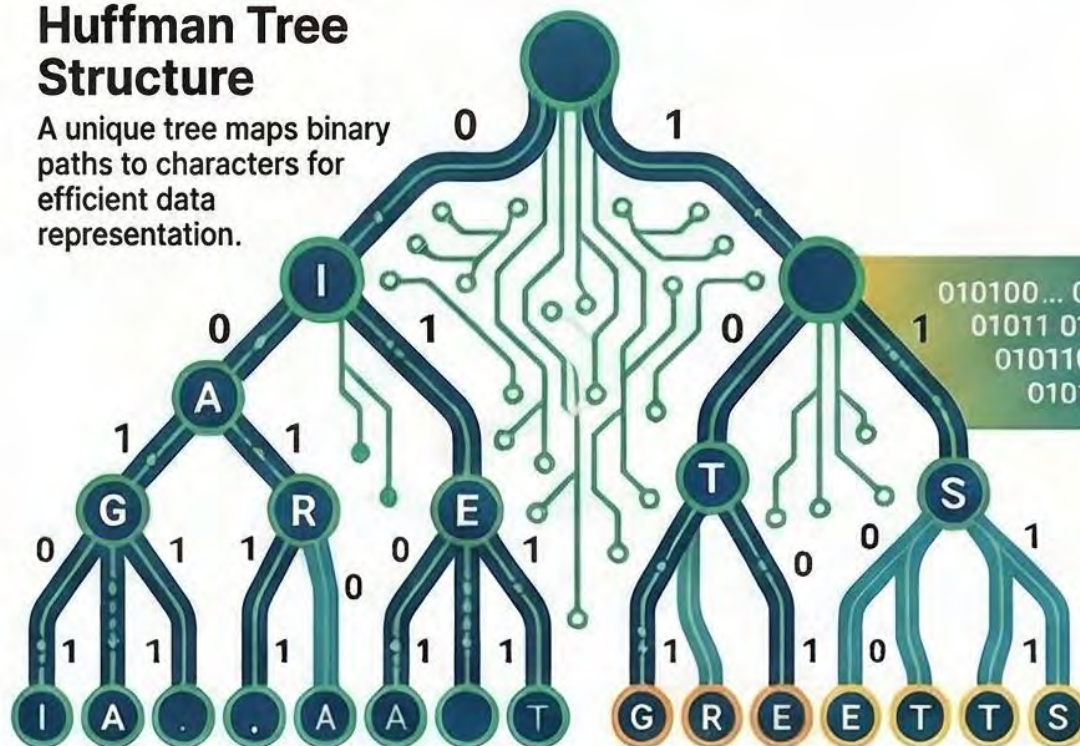
Frequency-Based Encoding

Characters in "INDIA IS GREAT" are assigned binary codes based on their frequency.



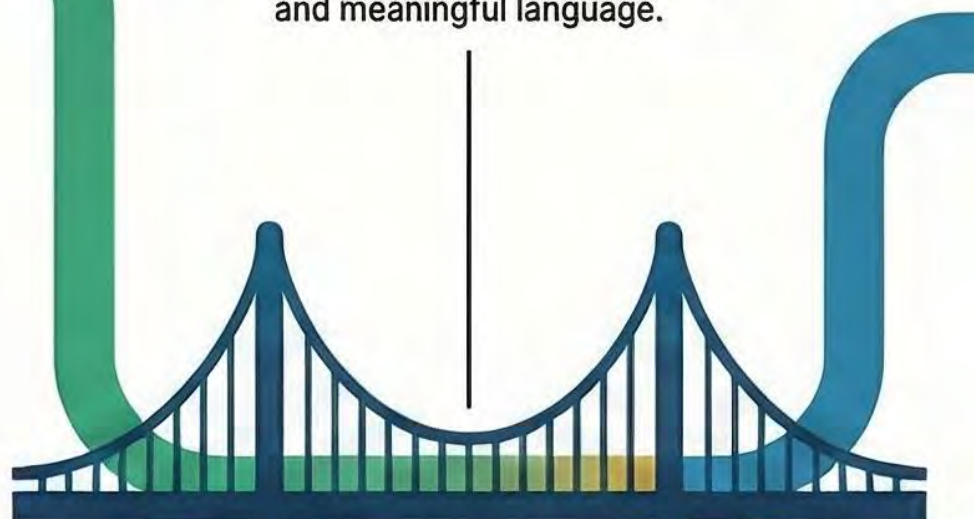
Huffman Tree Structure

A unique tree maps binary paths to characters for efficient data representation.

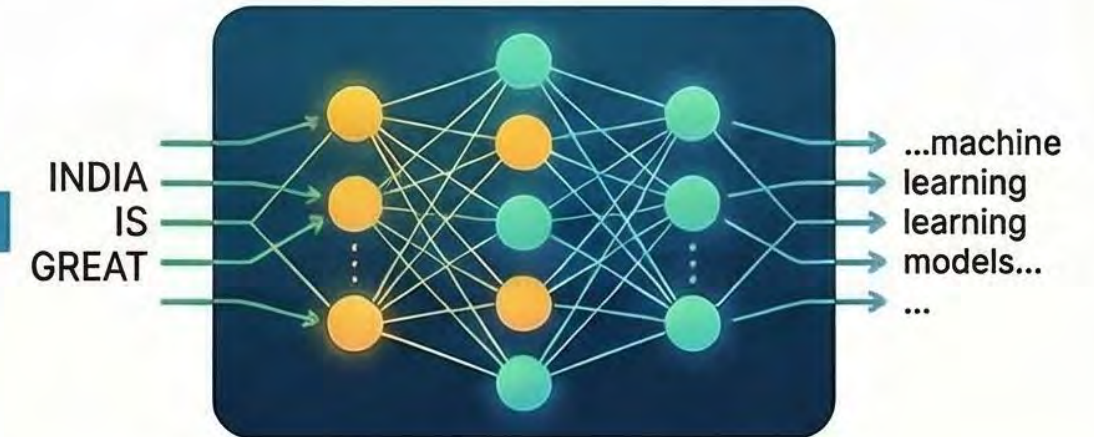


Tokenisation Bridge

Stored vocabularies act as the essential map between raw data and meaningful language.

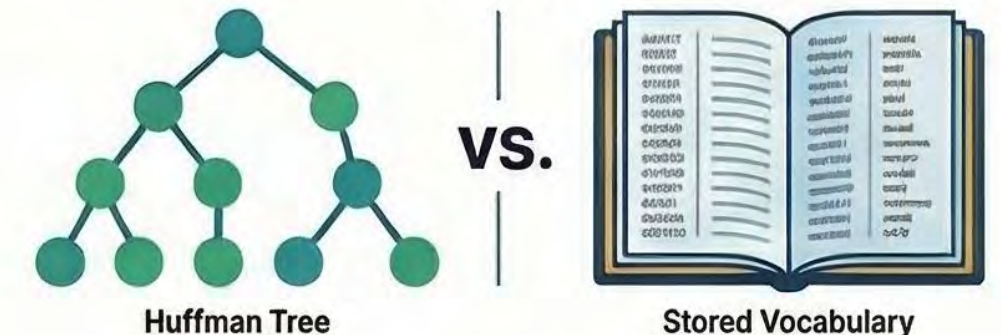


The LLM Analogy



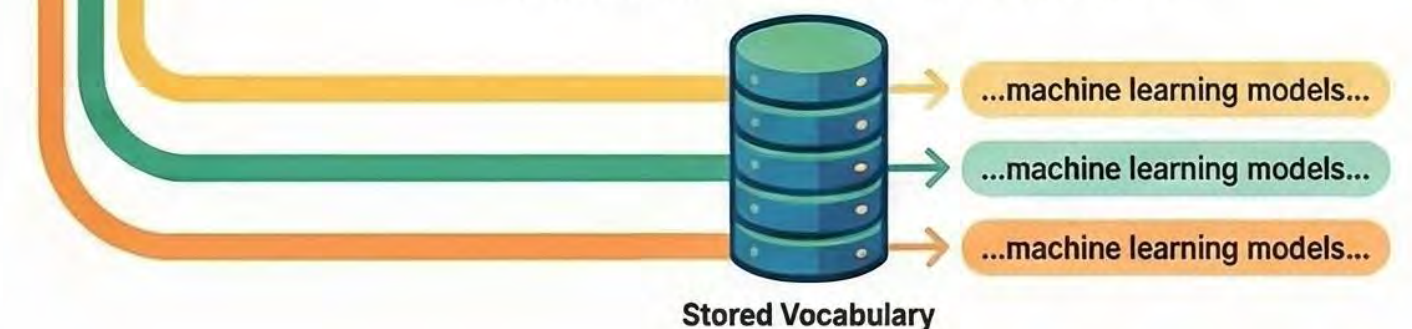
Trees vs. Vocabularies

Just as Huffman needs a tree, LLMs require a vocabulary to interpret tokens.



The Necessity of Storage

Models cannot convert numerical weights back into human-readable text without a stored reference.



The Mandatory Decoding Key

The Huffman Tree must be stored alongside compressed data to enable reconstruction.



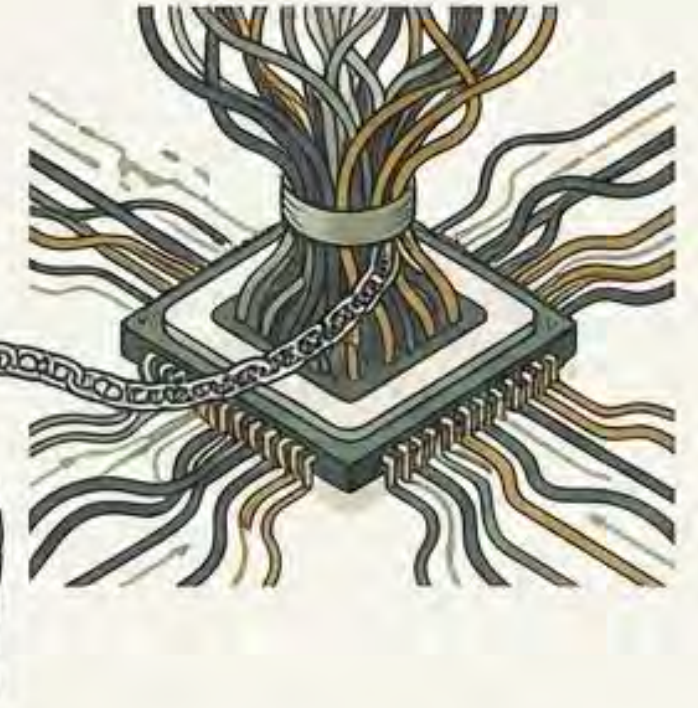
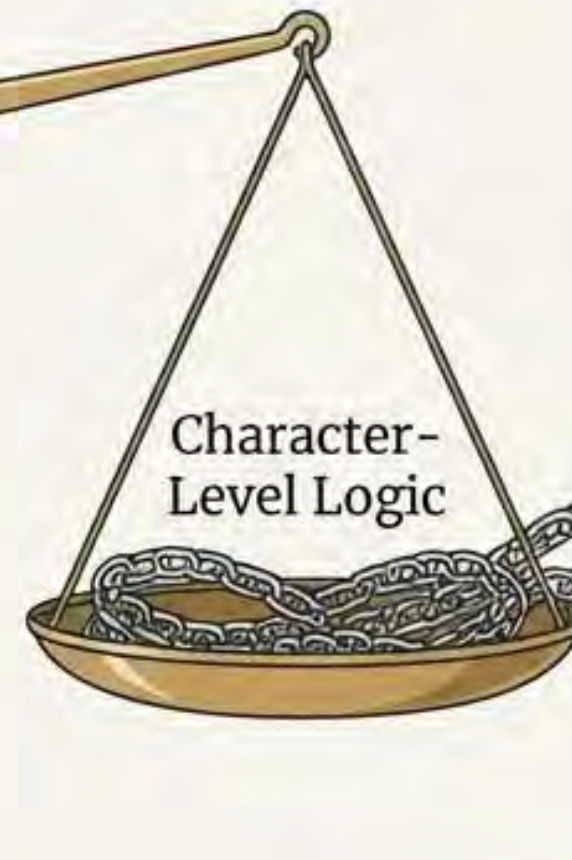
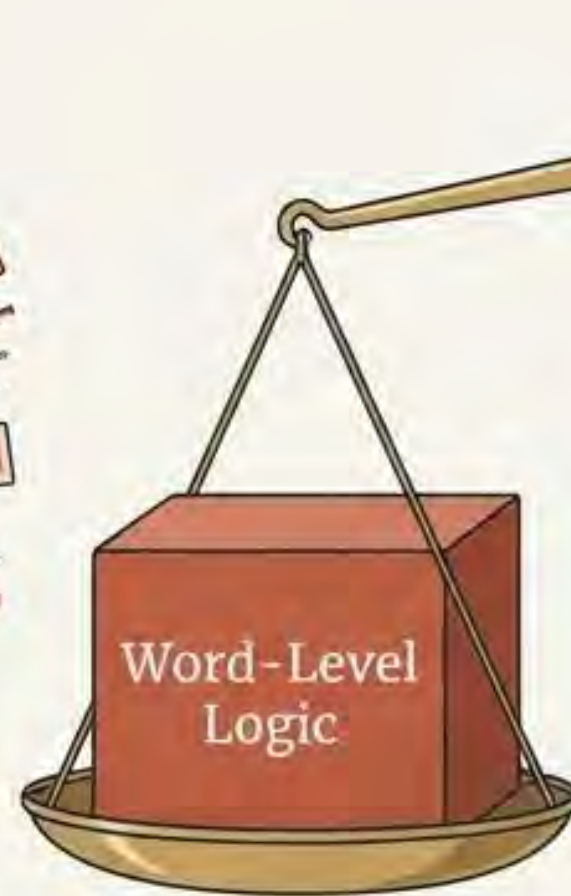
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The Foundational Mathematical Trade-off



Preserves deep semantic boundaries but induces an unmanageable explosion in vocabulary size (millions of unique tokens). Results in fatal Out-Of-Vocabulary (OOV) errors for novel terms.



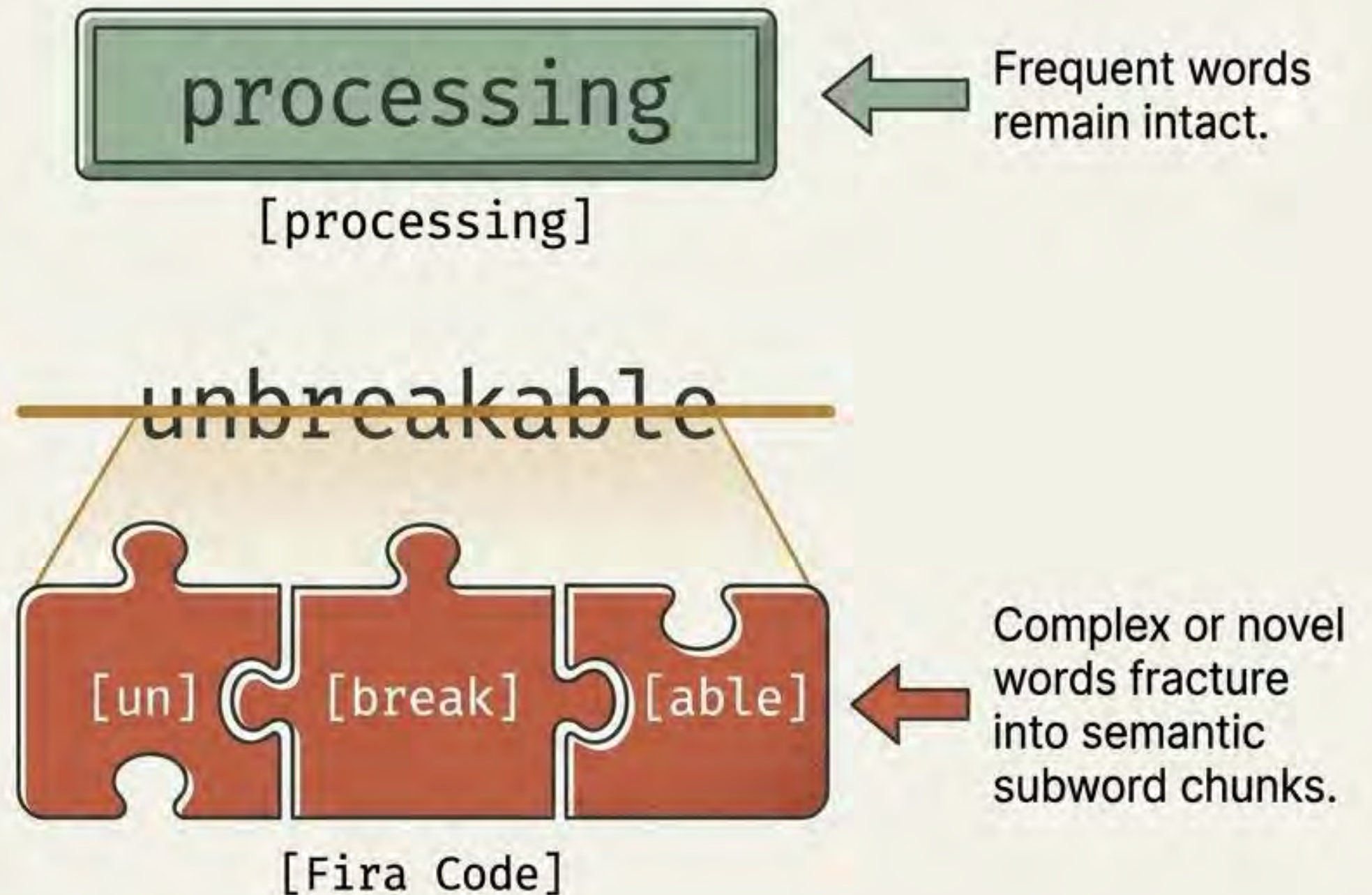
Collapses vocabulary to 100-256 tokens, guaranteeing zero OOV errors. However, produces sequences of paralysing length that instantly overwhelm the Transformer's $O(N^2)$ quadratic attention bottleneck.



The Subword Compromise

Modern neural architectures navigate the extremes using subword tokenisation, predominantly Byte-Pair Encoding (BPE).

By keeping frequent words intact and fracturing rare words into meaningful pieces, models balance a compact vocabulary with high semantic density.



The Empirical Baseline: Alice in Wonderland

A controlled empirical trial reveals profound structural divergences.

Despite a massive 50k+ vocabulary, the advanced GPT-2 BPE tokeniser produced a longer token sequence and a lower compression ratio than a naive whitespace splitter.

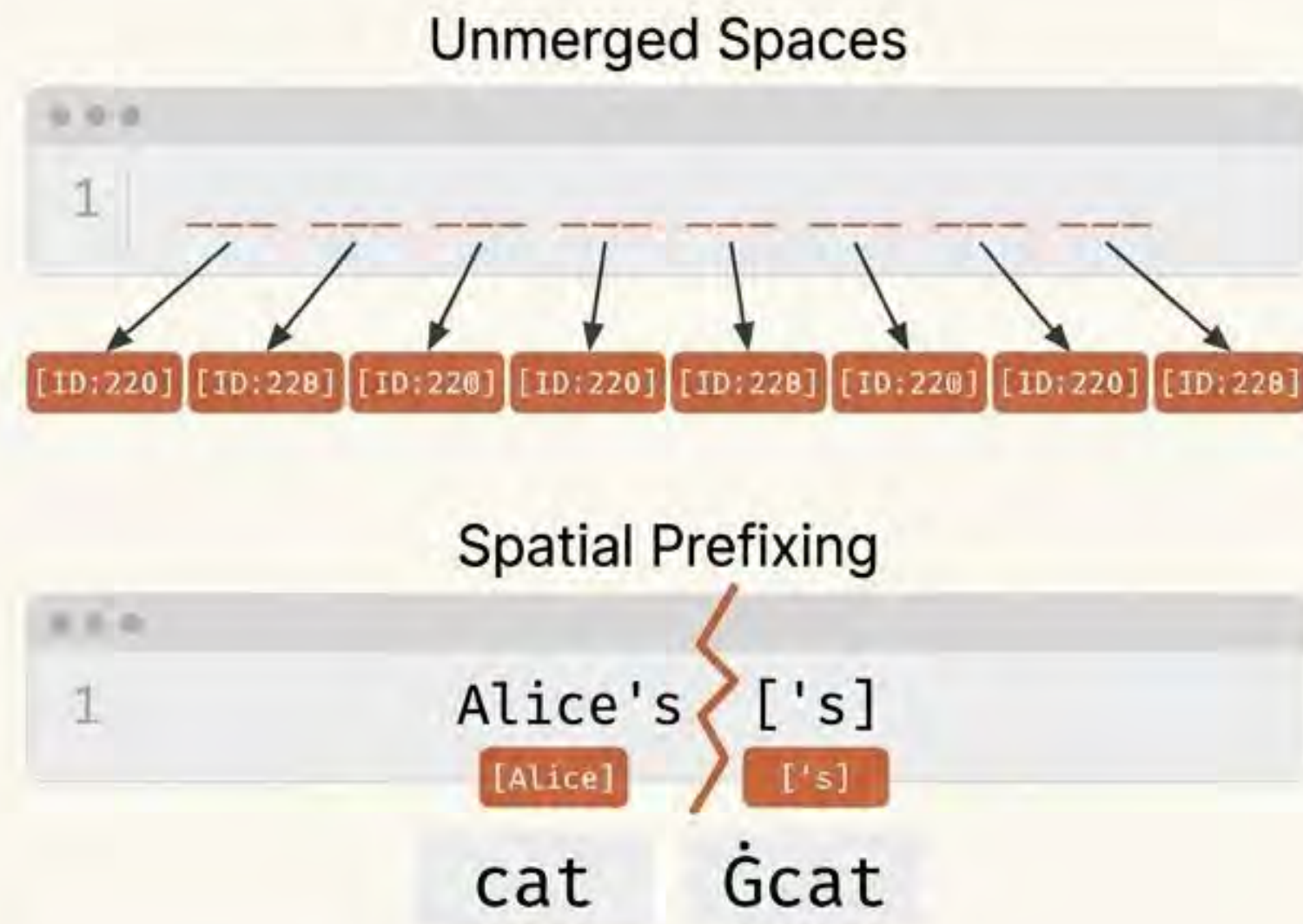
Tokenisation Strategy	Total Output Tokens	Vocabulary Size	Compression Ratio (chars/token)
Character-Level	167,713	93	1.00
Whitespace-Level	39,192	3,551	4.28
GPT-2 BPE (Subword)	52,951	50,257	3.17



Decoding the Whitespace Anomaly

GPT-2's compression deficit stems from hard-coded pre-tokenisation rules, not the algorithm itself.

- **Pathology of Unmerged Spaces:** Consecutive spaces were left unmerged, consuming massive context windows for simple formatting.
- **Spatial Prefixing:** Treating leading spaces as part of the token redundantly expands the vocabulary and aggressively fractures punctuation.



Insight

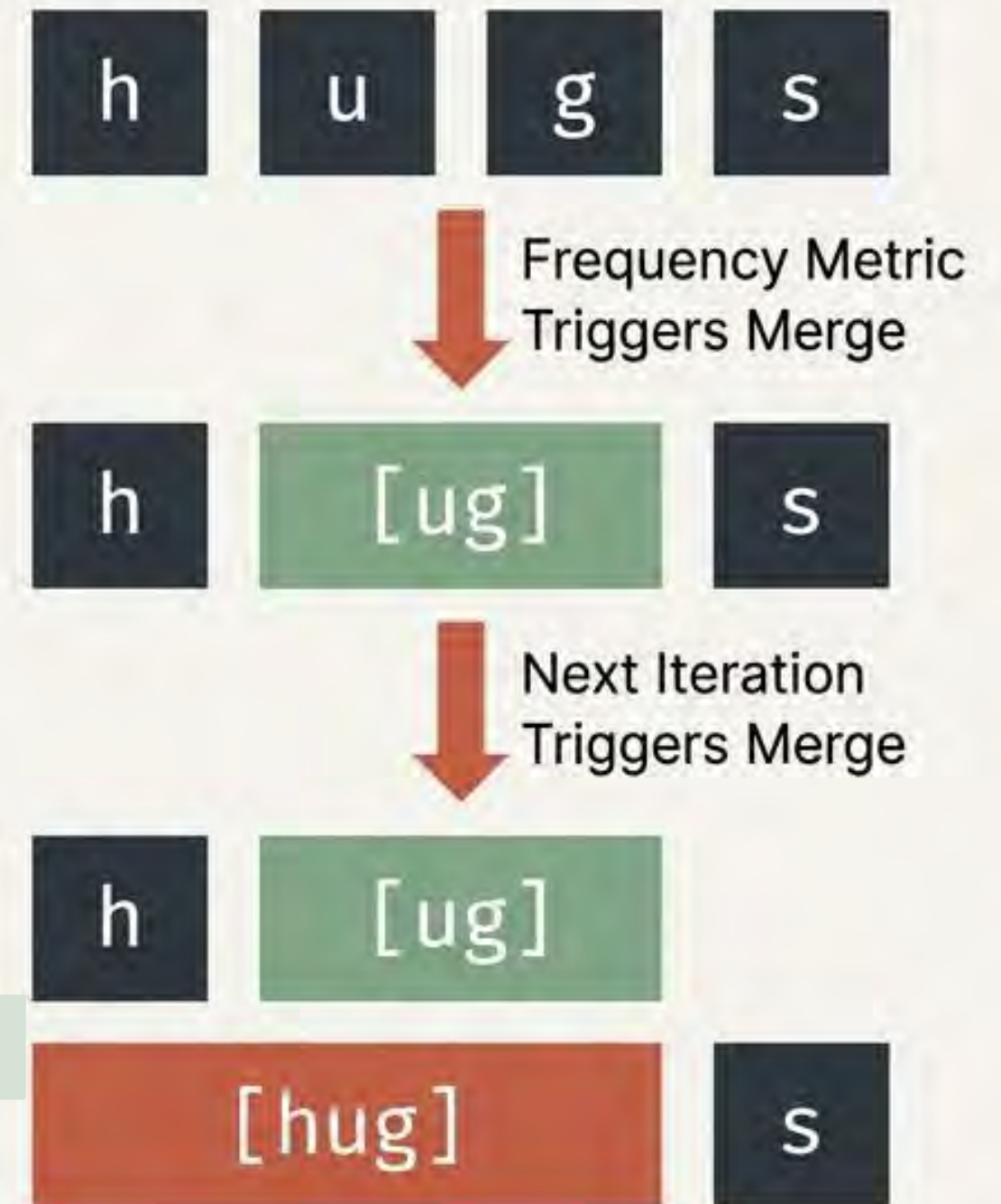
OpenAI explicitly fixed this flaw in GPT-4's cl100k_base vocabulary, aggressively forcing the aggregation of consecutive whitespace.

Anatomy of an Algorithm: Bottom-Up Merging

Byte-Pair Encoding (BPE) relies on purely deterministic, greedy frequency counting. It initialises with a base vocabulary and iteratively merges the most frequently co-occurring adjacent pairs until reaching a strict hyperparameter limit (e.g., 50,000 tokens).

Insight: Byte-Level

Insight: Byte-Level Preprocessing (BBPE). To guarantee zero OOV errors without needing an infinite Unicode vocabulary, OpenAI's models execute BPE merges directly on the 256 raw UTF-8 bytes.



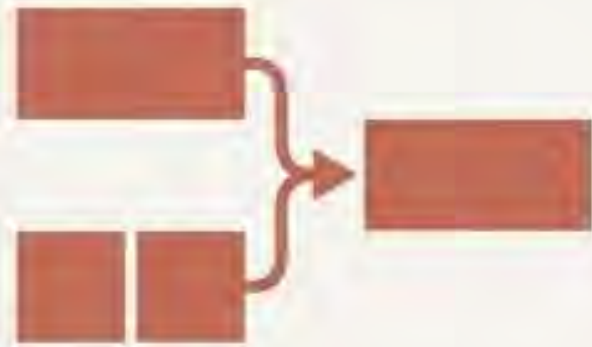


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The Diverse Subword Ecosystem

BPE (GPT, Llama)



Bottom-up greedy merging based entirely on co-occurrence frequency.

WordPiece (BERT)



Bottom-up merging based on probabilistic information gain. Privileges semantically meaningful chunks, using '##' for continuations.

Unigram



Top-down loss minimisation. Massively oversized initial vocabulary is pruned of tokens that contribute least to overall corpus likelihood.

SentencePiece



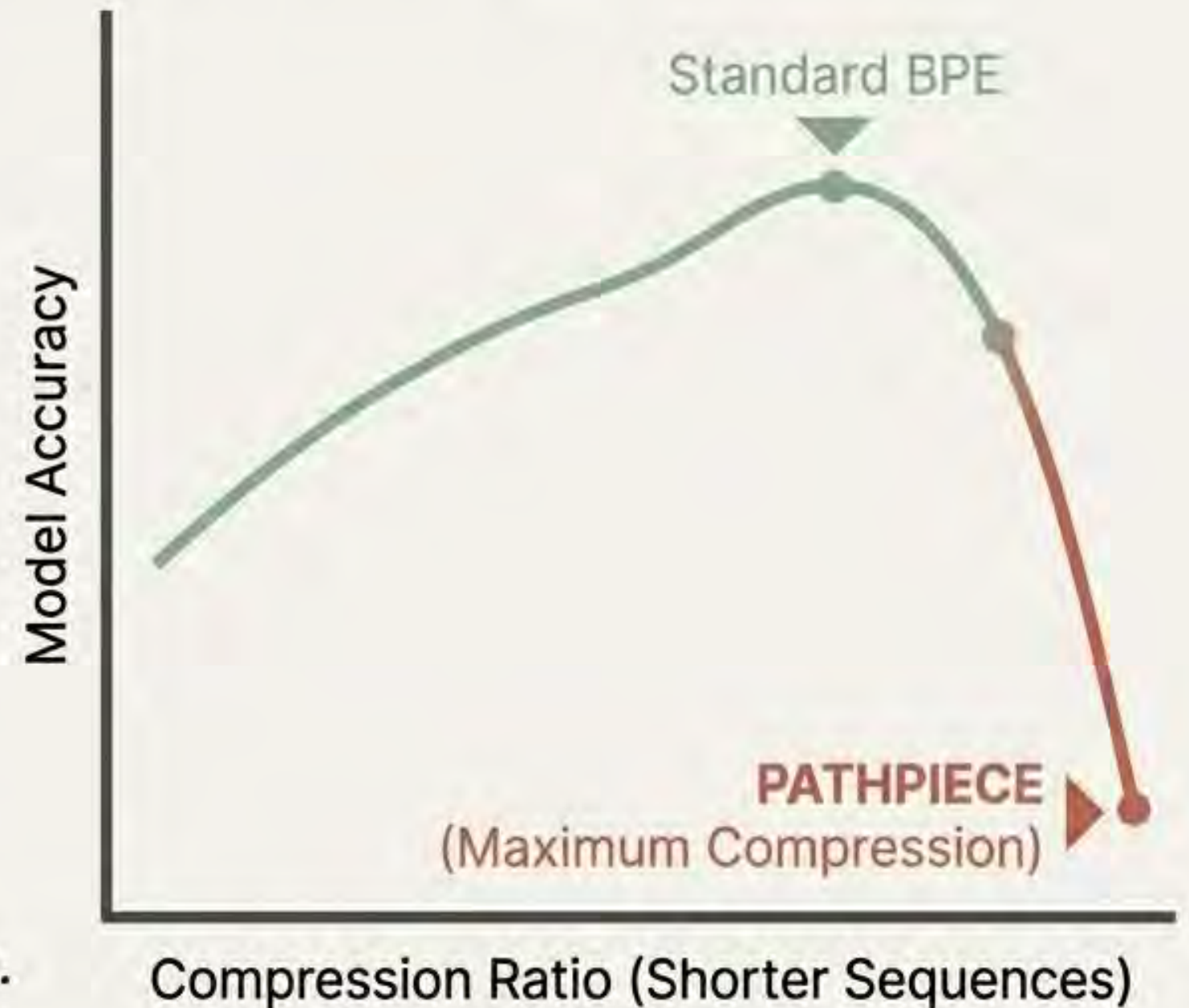
Bypasses the whitespace assumption. Treats current text as a raw Unicode stream, explicitly mapping whitespace ('_'), crucial for non-Latin scripts.



The Compresso Compression Fallacy

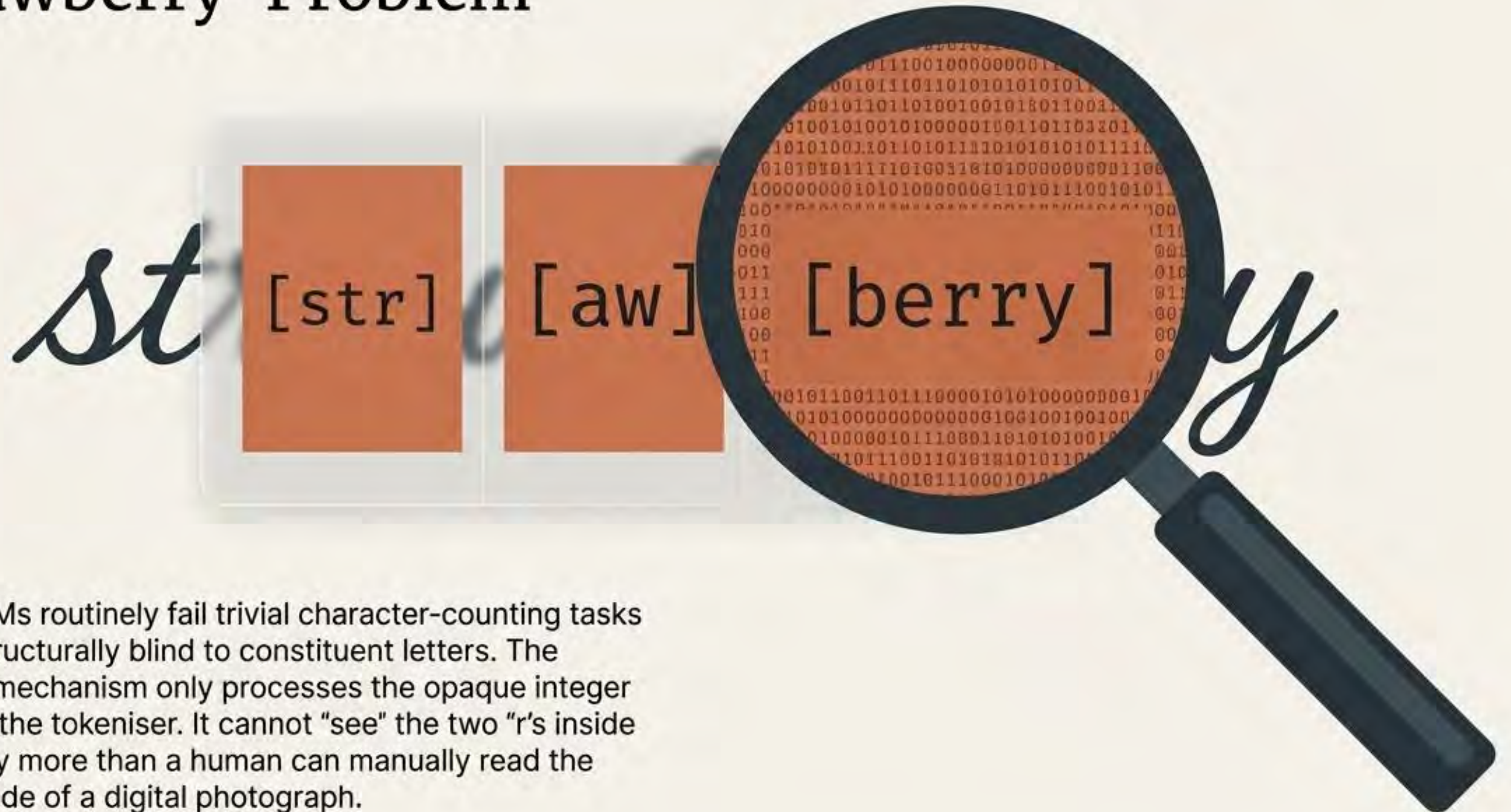
Tokenisation intersects heavily with classical information theory. But does maximum compression equal a smarter model?

- A 2024 Harvard/MIT study used PATHPIECE to guarantee the mathematical minimum sequence length for a corpus.
- **The Result:** Maximally compressed models consistently underperformed standard BPE.
- **The Reality:** BPE succeeds because its frequency-driven merges organically align with natural human morphological structures, not blindly optimised mathematical compression.





Cognitive Artefact 1: The 'Strawberry' Problem



Highly advanced LLMs routinely fail trivial character-counting tasks because they are structurally blind to constituent letters. The network's attention mechanism only processes the opaque integer chunks provided by the tokeniser. It cannot "see" the two "r's inside the token [berry] any more than a human can manually read the underlying binary code of a digital photograph.



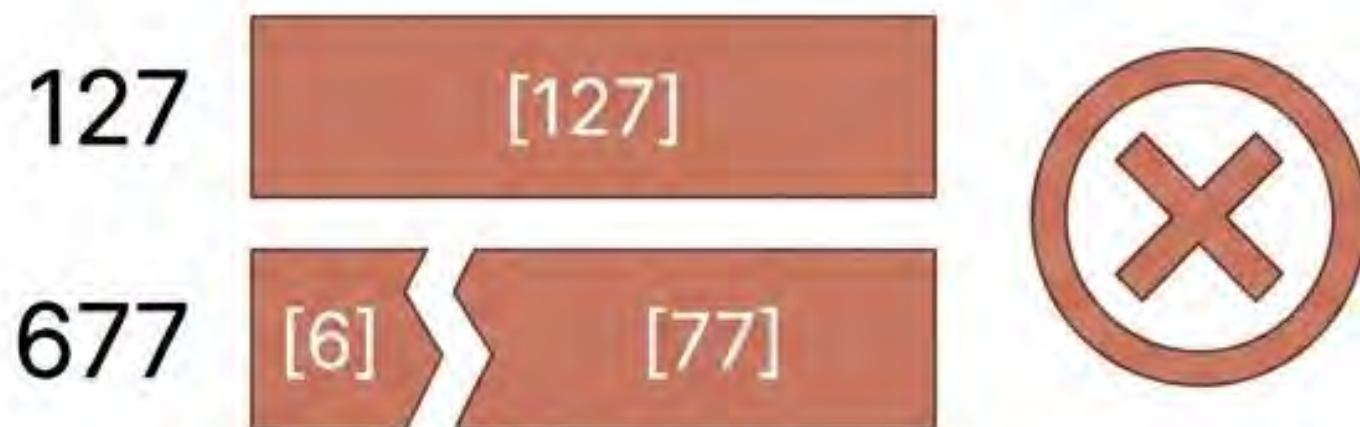
Cognitive Artefact 2: The Fragmentation of Math

Standard BPE treats numerical digits identically to text, merging them based on frequency.

This arbitrary fragmentation fundamentally destroys the mathematical place-value alignment within the transformer's attention window, crippling multi-digit reasoning.

Architectures like Meta's Llama resolve this by forcefully splitting all digits into single-character tokens.

Standard BPE



Llama 2/3 Fix





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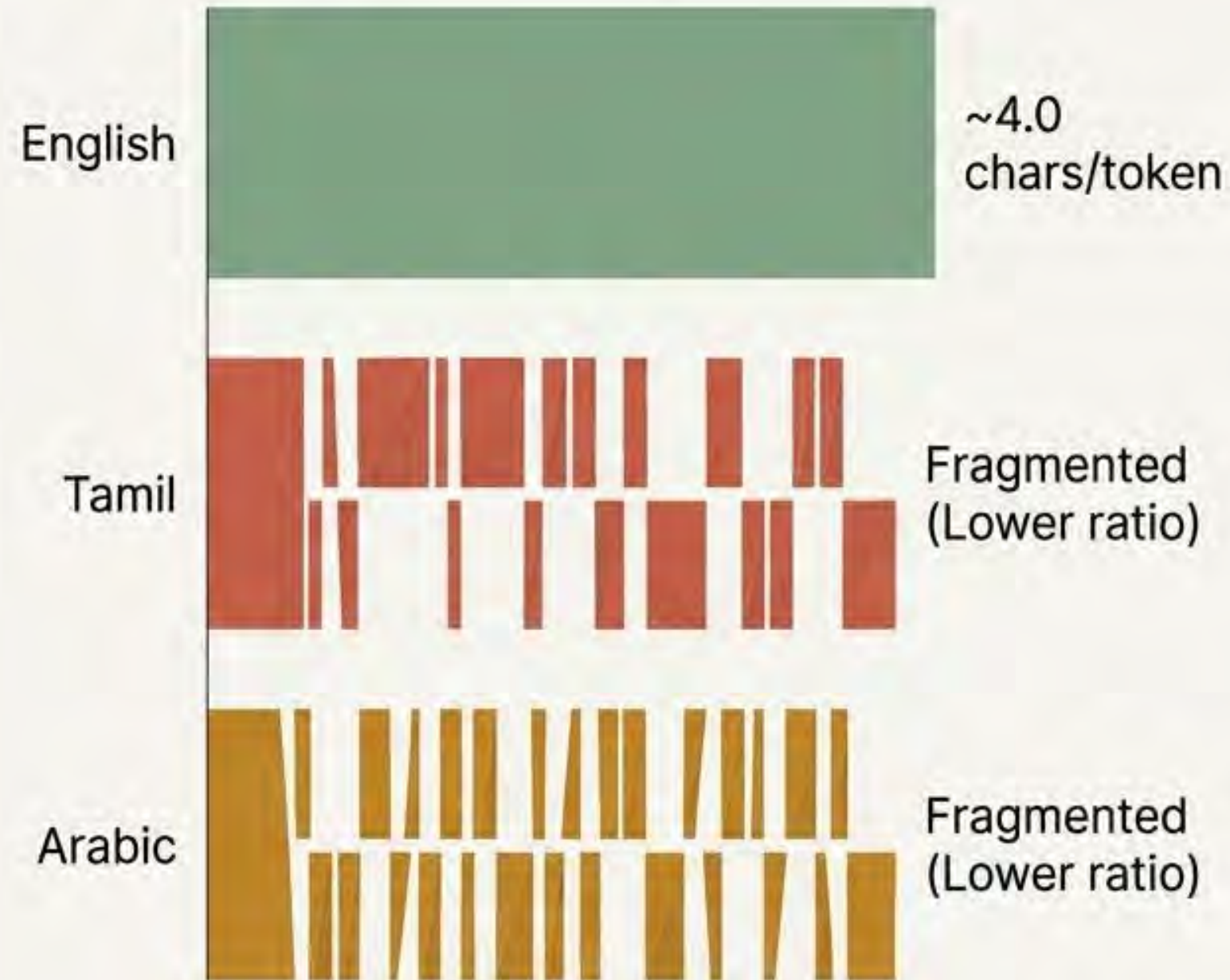
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The Multilingual Token Tax

Subword vocabularies are heavily biased towards their predominantly English training data.

While an English token often represents a complex, complete word, underrepresented languages are mercilessly fragmented into short byte sequences.

This introduces profound computational inequity to convey the exact same semantic information.



Slower Latency



Shorter Context



Higher API Cost



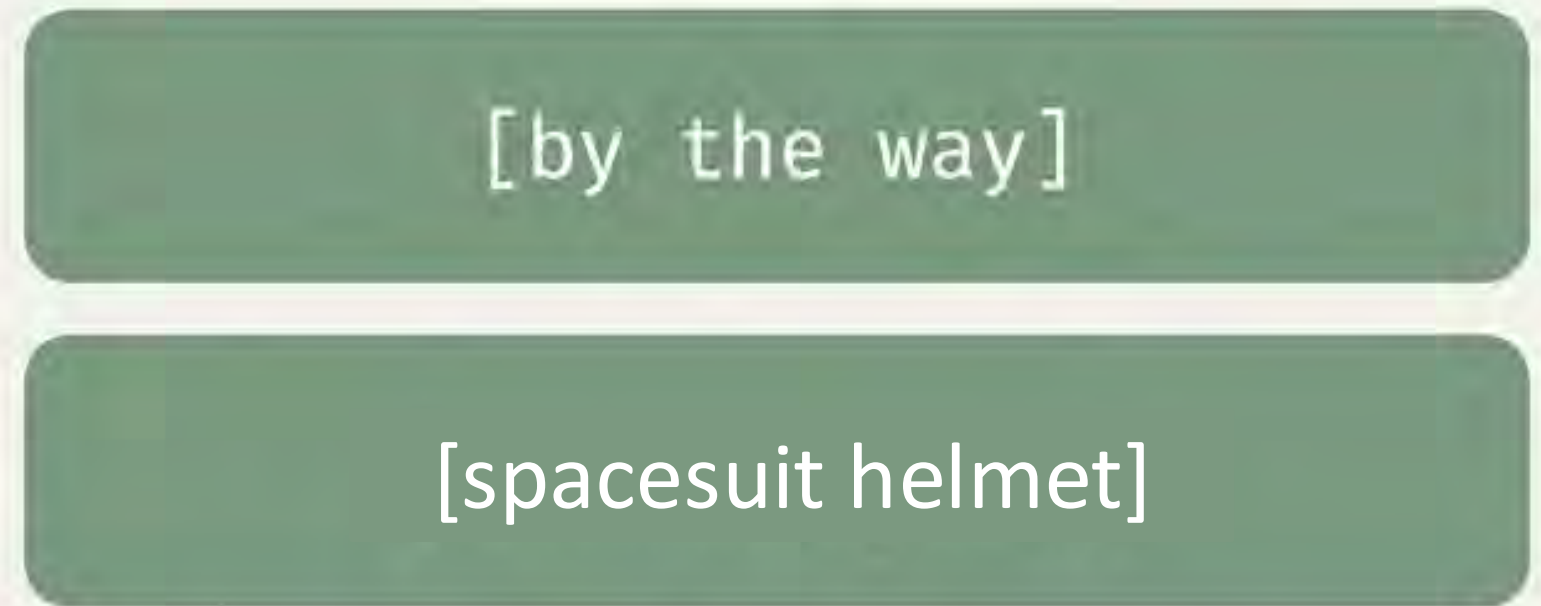
Architectural Frontier I: The Superword Paradigm

The traditional axiom that tokens cannot bridge whitespace is an arbitrary linguistic flaw. SuperBPE relaxes this constraint, identifying and fusing frequent multi-word expressions into single “superwords”.

Standard BPE



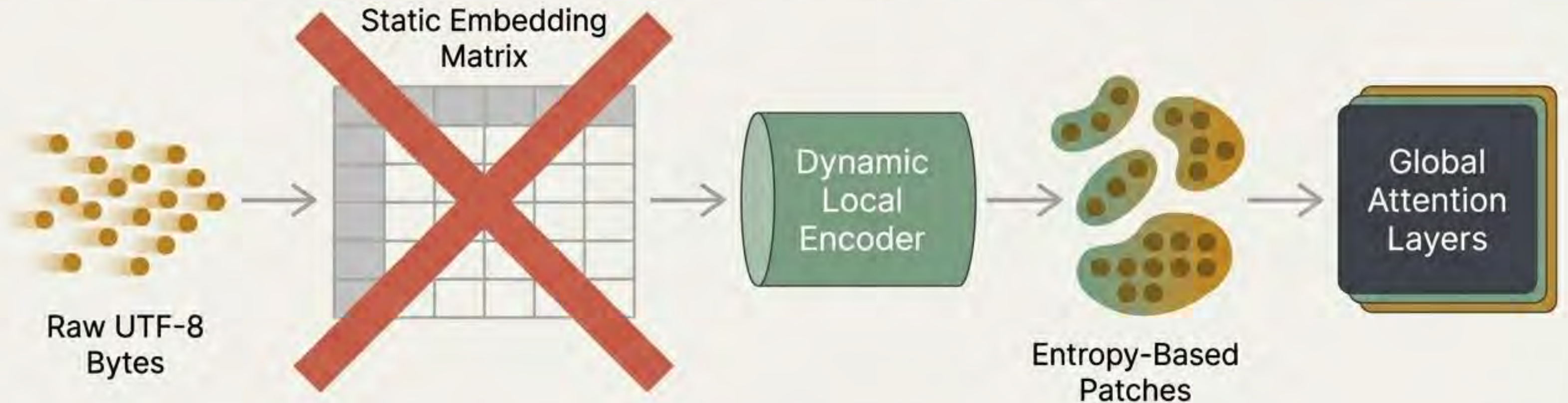
SuperBPE



The Impact: At a 200k vocabulary, SuperBPE encodes text with up to 33% fewer tokens. Pretrained 8-Billion parameter models achieve **27% less inference compute** while driving a massive **+8.2% absolute gain** on MMLU reasoning benchmarks.

Architectural Frontier II: Byte-Level Transformers

The most radical advancement involves dismantling the static tokenisation pipeline entirely. Byte-Level Transformers (BLT) ingest raw UTF-8 bytes directly.



To overcome the $O(N^2)$ bottleneck, a lightweight local encoder dynamically groups contiguous byte sequences into 'patches' based purely on internal data entropy. BLTs eliminate heuristic regex, OOV fragility, and the multilingual token tax, processing language with absolute lossless fidelity.



The End of Static Discretisation



Linguistic discretisation governs the fundamental capabilities of modern artificial intelligence. We have reached the physical and cognitive limits of the rigid Byte-Pair Encoding paradigm. Whether through whitespace-bridging Superwords that capture dense semantic intent, or Byte-Level Transformers that dynamically patch raw data streams, the era of the static vocabulary is closing. The future of text processing natively, flexibly, and equitably reflects the unbroken reality of human language.

Support Independent Research: Fueling Discovery

The Financial Barriers to Research



Essential Infrastructure Costs

Funds are required for software licenses, platform access, and database subscriptions.



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Contributions cover article processing charges, professional editing tools, and private reviewers.



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